# ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 6 | Carbon has 6 protons. |
| 2 | **(b)** ethane | Alkane. |
| 3 | **(d)** allotropes of carbon | Both are pure carbon. |
| 4 | **(a)** 4 | Tetravalent. |
| 5 | **(c)** ethene | Alkene. |
| 6 | **(b)** graphite | Graphite has free electrons. |
| 7 | **(d)** methane | $CH_4$. |
| 8 | **(a)** ethyne | Alkyne. |
| 9 | **(b)** diamond | Diamond. |
| 10 | **(b)** $CH_4$ | One C, four H. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Valency is 4 (1); It has 4 electrons in its outer shell (1). | 1+1 |
| 12 | Same molecular formula (1); Different structures (1). | 1+1 |
| 13 | Alkane $CnH_{2}n_{2}^{+}$ (1); Alkene $CnH_{2}n$ (1). | 1+1 |
| 14 | Compound of only carbon and hydrogen (1); $CH_4$ and $C_2H_4$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Methane $CH_4$ (1); Ethane $C_2H_6$ (1); Propane $C_3H_8$ (1). | 1+1+1 |
| 16 | Low melting and boiling points (1); Poor conductors of electricity (1); Mostly burn in air (1). | 1+1+1 |
| 17 | Sodium ethanoate and water (1); $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (2). | 1+2 |
| | **OR:** Ethene $C_2H_4$ (1); Propene $C_3H_6$ (1); Butene $C_4H_8$ (1). | 1+1+1 |
| 18 | Solvent (1); Fuel mixed with petrol (1); In medicines and tinctures (1). | 1+1+1 |
| | **OR:** Soaps: sodium/potassium salts of fatty acids (1); Detergents: sodium salts of sulphonic acids (1); Detergents work in hard water (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $CH_4$: 4 C-H single bonds (1); $C_2H_4$: 4 C-H and a C=C (1); $H_2O$: 2 O-H (1); Shared pairs: 4, 6 and 2 (1); Covalent bonds form by sharing (1). | 1+1+1+1+1 |
| | **OR:** $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Esterification needs conc. $H_2SO_4$ (1); Ester smells fruity (1). | 1+1+1+1+1 |
| 20 | Single bonds vs double/triple bonds (1); General formula differs (1); Saturated: clean flame (1); Unsaturated: sooty flame (1); Addition reactions in unsaturated (1). | 1+1+1+1+1 |
| | **OR:** Soap has two ends (1); Hydrophobic tail attaches to grease (1); Micelle forms (1); Hard water forms scum (1); Detergents solve this (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** saturated hydrocarbons | Single bonds only. |
| 2 | **(b)** combustion | Combustion. |
| 3 | **(d)** $C_2H_6$ | Two carbons single bond. |
| 4 | **(d)** propane | $C_3H_8$. |
| 5 | **(c)** $CO_2$ and water | Complete combustion. |
| 6 | **(b)** $C_2H_4$ | One double bond. |
| 7 | **(b)** -OH | Hydroxyl. |
| 8 | **(c)** alcohol | Spirit. |
| 9 | **(c)** $C_2H_2$ | One triple bond. |
| 10 | **(c)** -COOH | Carboxyl. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Alcohol (-OH) (1); Carboxylic acid (-COOH) (1). | 1+1 |
| 12 | Reaction of acid with alcohol (1); Gives ester and water (1). | 1+1 |
| 13 | Hydrocarbons with only single bonds (1); Ethane (1). | 1+1 |
| 14 | Diamond (1); Graphite (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Ethyne $C_2H_2$ (1); Propyne $C_3H_4$ (1); General formula $CnH_2n$-2 (1). | 1+1+1 |
| 16 | Vinegar (1); Food preservation (1); Making esters/dyes (1). | 1+1+1 |
| 17 | Insoluble precipitate in hard water with soap (1); Ca and Mg salts react (1); Reduces cleaning (1). | 1+1+1 |
| | **OR:** -OH alcohol (1); -CHO aldehyde (1); -COOH carboxylic acid (1). | 1+1+1 |
| 18 | $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (2); Heat and light released (1). | 2+1 |
| | **OR:** $CH_3{-}CH_3$ (1); $CH_2{=}CH_2$ (1); $CH{\equiv}CH$ (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Covalent bonding (1); Low melting/boiling points (1); Poor conductors (1); Burn in air (1); Form chains (catenation) (1). | 1+1+1+1+1 |
| | **OR:** Diamond: hard; cutting (1); Graphite: conducts; electrodes (1); Fullerene $C_{60}$ (1); Allotropes: same element, different structure (1); Diamond is an insulator (1). | 1+1+1+1+1 |
| 20 | Methane $CH_4$ (1); Ethane $C_2H_6$, propane $C_3H_8$ (1); Butane $C_4H_{10}$ (1); Pentane $C_5H_{12}$ (1); Hexane $C_6H_{14}$ (1). | 1+1+1+1+1 |
| | **OR:** Addition: unsaturated compounds add atoms (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (1); Substitution: H replaced by another atom (1); $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (1); Sunlight needed for substitution (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** ethanoic acid | Acetic acid. |
| 2 | **(c)** $C_2H_5OH$ | Alcohol with 2 carbons. |
| 3 | **(b)** -CHO | Formyl. |
| 4 | **(d)** fruity smell | Sweet smell. |
| 5 | **(a)** $CH_3COOH$ | Carboxylic acid. |
| 6 | **(a)** sodium salt of fatty acid | Soap structure. |
| 7 | **(a)** nickel | Ni. |
| 8 | **(b)** sharing of electrons | Sharing of pairs. |
| 9 | **(a)** $C_4H_{10}$ | Alkane $CnH_{2n+2}$. |
| 10 | **(c)** hard water | No scum. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Used for washing (1); Removes dirt and oil (1). | 1+1 |
| 12 | Contain double or triple bonds (1); Ethene (1). | 1+1 |
| 13 | Carbon atoms linking to form chains (1); Rings also form (1). | 1+1 |
| 14 | Carbon dioxide (1); Lime water turns milky (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Methanol (1); Ethanol (1); Propanol (1). | 1+1+1 |
| 16 | $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (2); Hydrogen gas evolved (1). | 2+1 |
| 17 | It is the hardest known substance (1); Each carbon bonded to four others (1); Rigid 3-D structure (1). | 1+1+1 |
| | **OR:** Methanoic acid (1); Ethanoic acid (1); Propanoic acid (1). | 1+1+1 |
| 18 | $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (2); Nickel (1). | 2+1 |
| | **OR:** Conducts electricity (free electrons) (1); Layers slide (1); Soft and slippery (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | n-butane (1); isobutane (2-methylpropane) (1); Structures drawn properly (1); Both are $C_4H_{10}$ (1); Pentane has three isomers (1). | 1+1+1+1+1 |
| | **OR:** Formula $C_2H_5OH$ vs $CH_3COOH$ (1); -OH vs -COOH (1); Ethanol neutral; acid turns blue litmus red (1); Only acid gives $CO_2$ with carbonate (1); Uses: solvent vs vinegar (1). | 1+1+1+1+1 |
| 20 | With Na: sodium ethoxide and $H_2$ (1); $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (1); With $KMnO_4$: ethanoic acid (1); With conc. $H_2SO_4$ at 443 K: ethene (1); Each reaction named (1). | 1+1+1+1+1 |
| | **OR:** Halo -Cl: $CH_3Cl$ (1); Alcohol -OH: $C_2H_5OH$ (1); Aldehyde -CHO: $CH_3CHO$ (1); Ketone C=O: $CH_3COCH_3$ (1); Acid -COOH: $CH_3COOH$ (1). | 1+1+1+1+1 |

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